UNIFORM VERTEX-PUSH CHAINS AS FOLDED-HYPERCUBE PRODUCTS: COMPONENT-ORDER INVERSION

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Abstract. Choose a vertex of an oriented finite simple graph uniformly and reverse every incident edge. On a push orbit, each connected component of order s is the labelled-generator walk on $\mathbb{F}_2^s/\langle \mathbf{1}\rangle \cong \mathbb{F}_2^{s-1}$: after a pivot is chosen, its generators are the coordinate vectors and the all-ones vector. Thus a nontrivial component is the standard folded-hypercube walk FQ_{s-1}, with an explicit duplicate-generator convention at $s = 2$; an isolate contributes an identity move. We assign this identification, the single-component folded- hypercube spectrum and bipartiteness, and established random-walk facts zero contribution credit. The disconnected chain is the degree-weighted random- scan product of these factors. Its multiplicity polynomial factors over component orders, which gives the all-time return formula. The residual result is an inverse for this product: when the ambient vertex number n is known, the transition spectrum determines every component order. A nearest-negative-root argument justifies a deterministic exact recovery algorithm whose only inputs are n and the spectral polynomial. Internal adjacency remains invisible; an unmarked transition kernel also contains no chosen starting orientation. Without known n, all positive-order edgeless graphs give the same one-state spectrum. No novelty or priority claim is made, and external circulation remains on hold.

1. Scope and owner subtraction

Pushing a vertex means reversing every incident edge. The operation and push-equivalence setting are established [2, 3], and Fourier diagonalisation of finite abelian Cayley walks is standard [4]. More specifically, the folded hypercube already has a Cayley presentation and complete character spectrum [6]; its bipartiteness boundary is recorded in the cycle literature [5], and random walks on this named graph family have been studied directly [1]. We therefore assign the vertex-push operation, the folded-hypercube identification below, the single-component spectrum and bipartiteness, generic Fourier and stationarity facts, and generic random-walk spectral moments zero contribution credit.

After that subtraction, the paper records two narrow items. First, it makes the labelled disconnected process explicit as a weighted random-scan product and packages its component factors in one multiplicity polynomial. Second, it proves and implements a known-n inverse that recovers the factor orders from that polynomial without access to the hidden component partition. This is an internal theorem record, not a novelty or priority claim; its status is `HOLD_EXTERNAL`.

Let $G = (V, E)$ be a finite simple graph with $n = |V| \geq 1$. Its connected components are $C_1, \dots, C_c$, of orders $s_1, \dots, s_c$. Fix an orientation ω_0. At every discrete time, choose $v \in V$ uniformly and push it. We restrict the chain (X_t) to one push orbit. Define

$$B_s(x) = \sum_{\substack{0\leq j\leq s\\ j \text{ even}}} \binom{s}{j} x^j, \qquad M_G(x) = \prod_{i=1}^{c} B_{s_i}(x). \tag{1}$$

2. Quotient and folded-hypercube product

Relative to ω_0, encode an orientation by the set of reversed edges. The mod-two cut map is

$$\delta : \mathbb{F}_2^V \longrightarrow \mathbb{F}_2^E, \qquad (\delta a)_{uv} = a_u + a_v \quad (uv \in E). \tag{2}$$

Lemma 2.1. *The push orbit is a torsor for*

$$A_G = \mathbb{F}_2^V / \operatorname{span}\{\mathbf{1}_{C_1}, \dots, \mathbf{1}_{C_c}\}, \qquad |A_G| = 2^{n-c}. \tag{3}$$

Pushes commute, the all-vertex push on each component is the identity, and these are the only relations. The orbit chain is irreducible and uniform measure is its unique stationary law.

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Proof. The condition $\delta a = 0$ says $a_u = a_v$ on every edge, hence a is constant on every connected component. Conversely, every componentwise constant vector lies in the kernel. Thus $\ker \delta = \operatorname{span}\{\mathbf{1}_{C_i} : 1 \le i \le c\}$. Pushing v adds δe_v, so the kernel gives all and only the relations and proves (3). The coordinate classes generate the quotient, and each translation is an involution. Irreducibility and uniform stationarity follow. These elementary orbit facts are setup rather than residual claims. $\square$

For $m \ge 2$, use the standard Cayley convention

$$FQ_m = \operatorname{Cay}(\mathbb{F}_2^m, \{e_1, \dots, e_m, \mathbf{1}_m\}), \qquad P_{FQ_m} = \frac{1}{m+1}\left(\sum_{j=1}^{m} T_{e_j} + T_{\mathbf{1}_m}\right), \tag{4}$$

where T_g denotes translation by g. This is the standard hypercube with complementary edges [6].

Proposition 2.2 (Exact product-kernel normal form)**.** *Choose a pivot $*_i \in C_i$. On the ith component quotient, the map*

$$\theta_i : \mathbb{F}_2^{C_i}/\langle \mathbf{1}_{C_i} \rangle \longrightarrow \mathbb{F}_2^{s_i - 1}, \qquad \theta_i([a]) = (a_v + a_{*_i})_{v \ne *_i} \tag{5}$$

is an isomorphism satisfying

$$\theta_i([e_v]) = e_v \quad (v \ne *_i), \qquad \theta_i([e_{*_i}]) = \mathbf{1}_{s_i - 1}. \tag{6}$$

Let $m_1 = |\{i : s_i = 1\}|$. On the product state space, the global Markov kernel is

$$P_G = \sum_{i : s_i \ge 2} \frac{s_i}{n} \left(P_{FQ_{s_i - 1}} \otimes \operatorname{Id}_{\ne i}\right) + \frac{m_1}{n} \operatorname{Id}. \tag{7}$$

Here $s_i = 2$ uses the convention described below.

Proof. Adding $\mathbf{1}_{C_i}$ changes every a_v and a_{*_i} simultaneously, so (5) is well defined. The representative with pivot coordinate zero proves bijectivity, and a direct substitution gives (6). For $s_i \ge 3$ the labelled generators are distinct and give exactly (4). A global step chooses component i with probability s_i/n and then one of its s_i vertex labels uniformly. Each isolate label acts trivially. Summing these mutually commuting factor kernels proves (7). $\square$

The low-dimensional conventions are essential. If $s = 2$, then the quotient has dimension one and the two labelled vertex choices both give the same nonzero translation. The unnormalised labelled operator is $2T_1$, while its normalised kernel is the usual two-state FQ_1 walk. If $s = 1$, the quotient is trivial and its sole labelled push is an identity move. Throughout, "transition spectrum" means the eigenvalues of the labelled-generator Markov kernel. It is not the simple-graph adjacency spectrum when duplicate generators or loops are suppressed. The identification in proposition 2.2 is claimed only as the explicit coordinate bridge to the owned folded-hypercube presentation.

3. Multi-component factorisation

The single folded-hypercube adjacency spectrum is known [6, Theorem 2.3]. In dimension $m = s - 1$, a character of weight r has unnormalised eigenvalue

$$m - 2r + (-1)^r. \tag{8}$$

For even $r = k$ and odd $r = k - 1$, this becomes the same value $s - 2k$. Pascal's identity combines the multiplicities to $\binom{s-1}{k} + \binom{s-1}{k-1} = \binom{s}{k}$, for even k. We use this single-component result as a zero-credit input.

Theorem 3.1 (Weighted-product spectrum)**.** *For $0 \le k \le n$, the transition eigenvalue*

$$\lambda_k = \frac{n - 2k}{n} \tag{9}$$

has multiplicity $[x^k]M_G(x)$. Consequently, for every integer $t \ge 0$,

$$\mathbb{P}(X_t = X_0) = 2^{-(n-c)} \sum_{k=0}^{n} [x^k] M_G(x) \left(\frac{n - 2k}{n}\right)^t. \tag{10}$$

The period is two exactly when every component order is even; otherwise it is one.

Proof. For component i, choose an even index k_i. Its unnormalised labelled eigenvalue is $s_i - 2k_i$ with multiplicity $\binom{s_i}{k_i}$; an isolate has only $k_i = 0$ and contributes its identity label. The weighted tensor sum (7) therefore has eigenvalue

$$\frac{1}{n}\sum_i (s_i - 2k_i) = \frac{n - 2\sum_i k_i}{n}.$$

Multiplying the independent component choices gives exactly the coefficient of x^k in (1). This proves the global factorisation.

Character orthogonality on the group of order 2^{n-c} gives the diagonal t-step kernel as the average of the tth powers of all transition eigenvalues, yielding (10). This is the standard spectral-moment consequence; folded-hypercube random walks themselves are not new here [1].

For period, if every s_i is even, each dimension $s_i - 1$ is odd and the corresponding folded cube is bipartite; equivalently, coordinate parity descends through every component relation and each labelled push flips it. Every return is even, while pushing one vertex twice gives a two-step return. If some s_i is odd, the all-vertex relation in that component is an odd closed walk, and the two-step return still exists, so the period is one. For $s_i = 1$, the odd walk is the isolate self-loop. The single-factor bipartiteness boundary is established background [5]; the statement here only records its random-scan product consequence. □

4. Known-order spectral inverse

The hypothesis that n is supplied is part of every statement in this section. Because the numbers $(n-2k)/n$ are distinct for different k, the known-n transition spectrum recovers every coefficient of M_G. Compress the even polynomial as

$$E_s(y) = \sum_{r\ge 0}\binom{s}{2r} y^r, \qquad Q_G(y) = \prod_{i=1}^{c} E_{s_i}(y), \qquad M_G(x) = Q_G(x^2). \tag{11}$$

Lemma 4.1 (Nearest-root separation)**.** *For $s \ge 2$, all roots of E_s are simple and negative. Its root nearest zero is*

$$\rho_s = -\tan^2\left(\frac{\pi}{2s}\right). \tag{12}$$

If $2 \le r < s$, then $|\rho_s| < |\rho_r|$ and $E_r(\rho_s) \ne 0$.

Proof. For real t,

$$E_s(-t^2) = \frac{(1+it)^s + (1-it)^s}{2} \tag{13}$$

$$= (1+t^2)^{s/2}\cos(s\arctan t). \tag{14}$$

The positive t-values producing roots are

$$t = \tan\left(\frac{(2j+1)\pi}{2s}\right), \qquad j \ge 0, \quad 2j+1 < s. \tag{15}$$

There are $\lfloor s/2\rfloor = \deg E_s$ distinct values, so their negative squares are all the roots and are simple. The case $j = 0$ gives (12). Since tangent is strictly increasing on $(0, \pi/2)$, $r < s$ gives $|\rho_s| < |\rho_r|$.

For the collision claim, put $t_s = \tan(\pi/(2s))$. If $E_r(\rho_s) = 0$, then (13) gives some $j \ge 0$ with

$$r\arctan t_s = \frac{(2j+1)\pi}{2}.$$

Here $\arctan t_s = \pi/(2s)$, hence $r/s = 2j+1$, impossible because the left side is less than one and the right side is at least one. Notice that this excludes the nearest root of the larger factor; it does not assert that every pair E_r, E_s is coprime. □

Theorem 4.2 (Input-only component-order recovery)**.** *Given only the positive integer n and the polynomial Q_G, the following deterministic exact procedure recovers the multiset of component orders. Starting with $R = Q_G$, scan $s = n, n-1, \ldots, 2$. While E_s divides R in $\mathbb{Z}[y]$, record s and replace R by R/E_s. After the scan, append*

$$n - \sum_{\text{recorded } s} s \tag{16}$$

copies of 1. The final remainder is 1, and the output is exactly $\{s_1, \ldots, s_c\}$.

Proof. Suppose the current remainder is a product of factors whose orders are at most s. If order s occurs, exact division removes one copy. Conversely, if order s does not occur but E_s divided the product of smaller-order factors, its nearest root ρ_s would be a root of at least one such factor, contradicting lemma 4.1. Repeated division therefore recovers the exact multiplicity of s. Descending induction proves that the scan removes all and only component orders at least two.

The loop declares a division successful only when its exact quotient lies in $\mathbb{Z}[y]$; division by a genuine remaining factor has this property because the current remainder is literally a product of the remaining E_r. The final remainder is 1, since $E_1 = 1$. Isolates leave no polynomial factor and are recovered precisely by the known total in (16). If $Q_G = 1$ initially, the procedure correctly returns n isolated vertices. □

The proof is root-based, but the executable recovery needs no numerical root finder: it uses exact integer polynomial division and the public inputs (n, Q_G) only. The hidden true partition is consulted solely after the routine returns, as a test oracle in finite controls.

5. Sharp inverse boundary and exact controls

Proposition 5.1. *The known-n transition spectrum does not determine internal component adjacency. A chosen starting orientation is not a target of this unmarked inverse. If n is not supplied, component orders are not recoverable in general.*

Proof. The factorisation (1) depends only on component orders. The connected graphs P_4 and K_4 have different internal adjacency but the same transition spectrum; the exact constructed kernels in the verifier both have characteristic polynomial $z^8 - z^6$.

Every push orbit is an affine coset of the cut image. Translation between any two cosets conjugates all labelled push transitions, and translation inside a coset changes the marked state. Hence neither the unmarked chain nor its spectrum records a chosen starting orientation or affine orbit. This is a category boundary, not an additional graph-reconstruction theorem.

Finally, for every positive ambient order, an edgeless graph has a one-state orbit and every labelled push is the identity. All such chains have spectrum $\{1\}$. Thus the known-n hypothesis cannot be deleted from a general component-order inverse. □

The deterministic standard-library verifier uses integers and `fractions.Fraction` only. Its round-one controls are summarized in table 1.

Table 1. Exact counterexample pressure; finite enumeration is not proof or ownership evidence.

control	coverage
labelled simple graphs, $1 \le n \le 5$	1,099
push-orbit states visited	14,149
return-recurrence state cells, $0 \le t \le 6$	71,874
fixed-total partitions and input-only recoveries, $n \le 30$	28,628
exact recovery divisibility attempts / successful peels	624,834/144,024
squarefree factors E_s, $2 \le s \le 30$	29
folded-quotient sizes including $s = 1, 2$	12
total exact assertions	155,901

For every graph, the program checks the complete cut kernel, orbit, labelled transitions, sign multiplicities, return recurrence, and period criterion. On every partition of each fixed total through 30, it constructs Q_G, calls the recovery routine with only (n, Q_G), and compares the returned multiset with the withheld partition. The former tautological root-order and known-factor-division controls have been removed. Exact gcd checks establish only squarefreeness through size 30; the analytic nearest-root order and collision exclusion are claimed only from lemma 4.1. Separately, the program builds the P_4 and K_4 edge sets, their labelled-generator transition matrices, and their exact characteristic polynomials. It also constructs affine-orbit and unknown-n edgeless witnesses.

No figure is included. The quotient map, weighted kernel, recovery algorithm, and root separator are exact formulas; a graph sketch or numerical root plot would add no evidence. The folded-hypercube inputs remain zero credit, the bounded owner search for the inverse is not novelty evidence, and external status remains `HOLD_EXTERNAL`.

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